

Alternative production of methanol from industrial CO₂

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ABSTRACT

Carbon dioxide valorization into value added products have become subject to much study to reduce industrial CO₂ emissions and fossil energy resource consumption. In this context, the purpose of this study is to evaluate and highlight the interest of CO₂ conversion into methanol through a complete techno-economic and environmental assessment of the entire process chain. The integrated process, successfully implemented in Aspen Plus®, is designed to treat the CO₂ coming from a conventional cement plant. A MEA-based CO₂ capture process is considered, and the captured CO₂ is then directly sent to the conversion unit for its catalytic conversion. Consequently, combining the two units leads to relevant integrations, especially regarding the reuse of the heat provided by the exothermal methanol reactions for the regeneration of the CO₂ capture solvent. An economic assessment is proposed to estimate the operational and investment costs, as well as the net present value, which demonstrates that the economic feasibility strongly depends on electricity and H₂ production costs. A Life Cycle Analysis method is finally performed to identify the main environmental hotspots. The underlying process design offers a significant reduction in greenhouse gases (besides other categories) when compared to the conventional fossil production from natural gas.

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1. Introduction

Nowadays, it is widely assessed that greenhouse gas (GHG) emissions are one of the most currently challenging environmental issues and that carbon dioxide is the largest anthropogenic GHG sources. GHG emissions have dramatically increased since the beginning of the industrial era: in 2018, the average concentration of CO₂ was 407 ppm [1], which is about 40% higher than in the mid-1800s, with an average growth of 2 ppm/year in the last ten years [2] and even 3 ppm/year since 2015 [3]. Among the human activities producing GHG, the combustion of fuels for electricity and heat is the largest source of CO₂ emissions representing more than 42% of the estimated anthropogenic CO₂ emissions, where 43% and 26% of these emissions are respectively dedicated to industrial and residential needs [2]. Apart from the energy generation, direct CO₂ emissions from the automotive transportation (23%) and industrial processes (19%) are also main contributors of the anthropogenic CO₂ emissions [2]. Furthermore, the global energy demand is expecting to increase throughout the years and even double by 2050 [4]. Fossil fuels utilization will thus remain predominant in

the energy sector in comparison to the use of renewable energy even if its penetration in the market is also increasing through time [4].

In this framework, Carbon Capture Utilization and Storage (CCUS) is the most commonly cited technique to reduce anthropogenic CO₂ emissions. However, Carbon Capture Storage (CCS) and Carbon Capture Utilization (CCU) techniques strongly differ from the absolute amount of CO₂ they can handle and their respective purposes need to be clarified. As presented in the IPCC mitigation scenarios, CCS applied to large CO₂ emitters (such as power, cement or steel plants) could jointly contribute to a CO₂ emissions reduction of above 25% by the year 2100 [5]. CCS is thus to be considered as a mitigation instrument for climate change even if the low current CO₂ tax restrain the creation of a business case for CCS technologies. On the other hand, CCU techniques propose to efficiently substitute the carbon raw material used in chemical processes by CO₂ in order to produce fuels or high-added value chemicals [6]. However, CCU technologies are sometimes considered as technologies that do not allow the reduction of CO₂ concentrations in the atmosphere [5,7]. Indeed, they generally do not allow storage of CO₂, as the CO₂-based product will emit back its incorporated CO₂ into the atmosphere after a short period (e.g. liquid fuels). Carbon mineralization is mostly the only CCU technique that can be considered for long-term CO₂ sequestration having a direct impact

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on climate change mitigation. Consequently, CCU potentials in reducing CO₂ emissions is rather low and even highly optimistic estimates currently assume that the total amount of CO₂ possibly used for CCU would reach about 180 Mt for chemicals and eventually 2 Gt for fuels, compared to the 37 Gt of anthropogenic emitted CO₂ [5]. As a result, CCU techniques will not contribute significantly to the mitigation of climate change but have to be considered as a strong component in a larger mitigation strategy plan including the reduction of the use of fossil fuels.

To this extent, innovative chemical processes are studied for years to propose efficient alternatives that are able to absorb CO₂ as carbon raw material sources and to convert/store it in chemically valuable compounds. Power-to-Liquid (PtL) systems have been envisaged so far as one of the most promising technology to answer this problematic by converting industrial CO₂ into liquid fuels that are more convenient to envisage for transportation, safety and infrastructure considerations [4]. As a result, one of the most promising fuel is methanol due to its liquid state at ambient conditions and its wide range of applications in the chemical and energy sectors. Indeed, methanol is also one of the most important building blocks of the chemical industry being used as feedstock for the synthesis of acetic acid, MTBE, DME and formaldehyde, which are also used in products like adhesives, foams, plywood subfloors, solvents and windshield washer fluid, etc. As also presented by Nobel Prize George A. Olah [8], methanol can also substitute oil derivatives and be directly used as fuel for heat engines and fuel cells due its high octane rating allowing higher compression ratio and more efficient combustions [4]. All these trends considered, the global methanol market is projected to expand and reach more than 48 billion euro by 2021, registering a Compound Annual Growth Rate (CAGR) of 12.4% between 2016 and 2021 [9].

The methanol formation is an exothermal reaction with a reduction of reaction mole number and is thus favored by temperature decrease and pressure increase. Methanol can be produced on a large scale process based on the following reactions: methanol synthesis from CO (1), Reverse Water Gas Shift (RWGS) (2) and methanol synthesis from CO₂ (3) [10]:



Investigating the hydrogenation possibilities of CO/CO₂, first studies of the CO/CO₂-based methanol synthesis were carried out as early as the mid-90s and described the catalytic and reaction system in terms of thermodynamics, reaction kinetics, catalysts development and scale up [11–13]. More recently, the potential of use of crude CO₂, as an alternative feedstock replacing CO in the methanol production, has received attention as an effective way of CO₂ utilization [14]. Even if the reactions and mechanisms occurring in the methanol production from CO₂ mixtures have been studied through the last decades, it is still difficult to make a reliable choice between various chemical equilibrium and kinetic constants as literature data are numerous [15–20] and generally conflicting. Graaf et al. [15–17,21] presented a rather complete study of the reactional mechanisms in terms of equilibrium and kinetics on a commercial CuO/ZnO/Al₂O₃ catalyst. The kinetics were also confirmed by An et al. [22], Lim et al. [23] and Kiss et al. [24] and will be thus considered in this work.

Since the 1920s, the main way to produce methanol uses syngas made from the steam reforming of natural gas in one-step, two-step or autothermal reforming processes [25]. Due to the high amount of fossil resources required for its synthesis, the typical CO₂

emissions for the methanol synthesis from syngas is 0.5 ton CO₂-eq/ton methanol but much higher values can be found when envisaging its synthesis from partial oxidation of residual oil, where the emissions are estimated to 1.4 ton CO₂-eq/ton methanol [26]. The average CO₂ emissions of European methanol plants is estimated to 0.76 ton CO₂-eq/ton methanol due to the use of both these technologies [4,26,27]. The direct CO₂ hydrogenation to methanol has been investigated in several Life Cycle Analysis (LCA) studies as reported by Artz et al. [27] to evaluate its impacts on different environmental metrics including Global Warming, Fossil Depletion and Human Toxicity. They indicate that these impacts can be lowered in comparison to the conventional production of methanol from syngas. However, this consideration is strongly depending on the energy sources (heat and electricity) considered in the studies as it largely influences the environmental evaluation of the process. A comparative environmental assessment between the proposed alternative to the conventional approach is thus mandatory to ensure its sustainability and its positive contribution to the climate change mitigation.

As a direct consequence of the industrial interest regarding alternative production of methanol, the study of the methanol synthesis from purified industrial CO₂ is thus a trendy topic. Various studies have already been performed regarding this conversion showing the theoretical feasibility of the process [4,26–30]. They are often combined with an economic assessment pointing out that variable operating costs are mostly dedicated to the production of the hydrogen [26] required for the hydrogenation of CO₂. Some studies are also focusing on the optimization of the conversion process itself presenting various heat integrations inside the methanol conversion unit [4,26,31]. In particular, Van-Dal et al. [32] considers the possible heat integration between the CO₂ capture and conversion units. Finally, the economic balance of the process is directly linked to its operative costs and especially the cost of electricity. As a result, the possible profitability of the overall installation is very sensitive because the cost of electricity can vary from 30 €/MWh [4] to 70 €/MWh [33] according to the economic scenario considered in the various studies. Moreover, care should be taken to consider the price of electricity related to renewable energy, mandatory to keep a positive LCA index regarding the overall process.

Consequently, this work aims to provide an original integrated process to capture CO₂ from industrial flue gases and to convert it into methanol reducing both the CO₂ emissions and the consumption of fossil resources dedicated to the methanol synthesis. In the studied process, methanol is produced from purified carbon dioxide coming from cement plants, widely recognized as the largest industrial CO₂ emitter, especially due to the decarbonation reaction (two thirds of the released CO₂ emissions), specific to the clinker production process [34,35]. The CO₂ capture process considered in this work is the amine-based absorption-regeneration process using a 30 wt% monoethanolamine (MEA), developed at industrial scale and widely considered as the benchmark for the purification of large industrial CO₂ emitters [36].

The original aspect of this work is to consider the entire CO₂ conversion chain from its capture to its conversion into methanol and to present the economic and environmental benefits of the integrations between the CO₂ capture and conversion units. These integrations comprise (i) the reuse of the heat provided by the exothermal methanol conversion reactions for the distillation of the methanol-water product and for the regeneration of the CO₂ capture solvent; and (ii) the use of a fraction of the water produced during the methanol synthesis to the water makeup of the CO₂ capture unit. To demonstrate the benefits from these integrations, economic assessments are provided to estimate the operational (OPEX) and investment (CAPEX) costs of both processes. A LCA is

also presented to highlight the environmental benefits of the integrated process in comparison to (i) the non-integrated process, and (ii) to the conventional methanol conversion process from syngas produced by steam methane reforming (currently representing 90% of the methanol production).

2. Methods

The alternative production of methanol from industrial CO₂ was evaluated through the following 3E performance indicators (Engineering-Economic-Environmental). They were divided into several metrics to assess the different aspects of the CO₂ capture unit and the CO₂ conversion unit.

2.1. Engineering performance

The technological indicators were direct results from the process modeling using the engineering software Aspen Plus® v10.

The mass balance metrics assess the mass demand of individual inputs and outputs, including the total mass of CO₂ and H₂ converted (see Eq. (4) and Eq. (5), respectively).

$$CO_2 conv = \left(\frac{CO_2(in) - CO_2(out)}{CO_2(in)} \right)_{process} \quad (4)$$

$$H_2 conv = \left(\frac{H_2(in) - H_2(out)}{H_2(in)} \right)_{process} \quad (5)$$

The energy balance metrics evaluate the utilities demand, including heat and electricity duties, and assess the possibility of energy integration. Systematic process-to-process heat recovery through data evaluation, pinch analysis, and optimized heat exchanger network is performed on the systems to reach high energy efficiency and minimize utility costs.

2.2. Economic performance

The economics of the process are based on both the capital investment (compressors, heat exchangers, reactors, flash tank, distillation column, etc.) and the operative expenses (compressor works, reboiler duty, catalyst change, etc.).

Capital Expenditure (CAPEX). Aspen Plus Economics® software was used to calculate the CAPEX of the installation. The calculation is based on costs for a variety of pieces of equipment and utilities, and equations for their correction factors.

The equipment costs usually comprise two different contributions, the purchased equipment cost, only considering the purchase costs of the equipment; and the installed equipment cost, considering the purchase and installation costs of the equipment. A quite extensive set of correlations is available in Guthrie [37], Douglas [38] or Peters and Timmerhaus [39] for the calculations of purchased equipment costs, which can be estimated using Eq. (6).

$$Purchased\ Cost = (Base\ Cost)(Index)F_c \quad (6)$$

Where F_c corresponds to a correction factor (e.g. for materials, pressure, etc.). The index evaluates the cost increase of equipment with time, the Marshall and Swift (M&S) index of August 2018 is used in this work; this index is monthly updated in the *Chemical Engineering Journal* [38].

Regarding the installed equipment costs, they are related to the purchased equipment costs by the addition of an installation factor IF (see Eq. (7)):

$$Installed\ Cost = (Base\ Cost)(Index)(F_c + IF) \quad (7)$$

This installation factor is also tabulated according to the equipment considered in Guthrie [37], and other sources.

Operative Expenditure (OPEX). For the calculation of the OPEX, the following values are considered: electricity costs of 70 €/MWh [33], catalysts costs of 10 €/kg [40], MEA costs of 1.03 €/kg [41], steam costs of 30 €/MWh, oxygen selling price of 54 €/ton O₂ [42] and CO₂ credit tax of 20 €/ton CO₂ (CO₂ European Allowances in August 2018) [43] and methanol selling price of 450 €/ton CH₃OH (Methanex non-discounted reference price for 2018) [44]. The lifetime of the catalyst has also been estimated to 3 years for CAPEX estimations [45] and end-life treatments can be neglected from an economical point of view as only metal recovery could be applied to this catalyst due to the different deactivation processes occurring during the methanol synthesis reactions [46]. The costs related to the dismantling of the infrastructure are not considered in this work.

Profitability. The calculation of the Net Present Value (NPV) is a conventional financial criterion to assess the profitability of the CO₂ capture and conversion process. This criterion is calculated as following (Eq. (8)):

$$NPV(i, N) = \sum_{t=0}^N \frac{R_t}{(1+i)^t} \quad (8)$$

Where t is the time of the cash flow, i the discount rate (i.e. 8%) [42], N the number of periods and R_t the net cash flow.

2.3. Environmental performance

LCA is achieved to evaluate the environmental performance of both processes to identify the environmental hotspots (i.e. processes or flows responsible for a significant share of the overall impact) and compare their respective impacts. Processes are therefore assessed using the results of process modeling from Aspen Plus® v10 and LCA formulation from SimaPro® v8.3 modeling software coupled with the Ecoinvent v3 database [47] to model the process and quantify the materials and energy supplies, choosing economic value as allocation criteria. The Aspen Plus® software is therefore used for the Life Cycle Inventory (LCI) data for the conversion processes that result from process modeling while the Ecoinvent database is used to provide background process LCI data.

The LCA applied is a gate-to-gate analysis. The system boundary comprises CO₂ capture and conversion processes as well as the hydrogen production process. The suggested design, involving integrations between both CO₂ capture and CO₂ conversion units, especially regarding the reuse of the heat provided by the exothermal methanol reactions for the regeneration of the CO₂ capture solvent, and the reuse of water as coproduct of the methanol synthesis for the water makeup of the CO₂ capture unit, constrains the distance between both units. As the result, all processes are considered on-site thus neglecting the environmental impacts and energy requirements of transports and storage. The production of one ton of methanol is considered as functional unit. For both CO₂ capture and conversion processes, the European energetic mix ENTSO-E is chosen for the electricity supply and considers the electrical mix from 2016: 42.2% fossil fuels, 33% renewable, 22.5% nuclear and 2.3% others net generation [48]. Steam requirement is taken from Ecoinvent database, assuming 76% natural gas and 24% oil as feedstock. Regarding the CO₂ conversion process, the catalyst is modelled using the information available for the manufacturing of chemicals from Ecoinvent database [49]. The spent catalyst, its recovery and further reuse are not included in the system boundary. As a result, no emission or credit was accounted for the spent catalyst. Regarding the hydrogen supply, various studies [26,50,51]

highlight that hydrogen must be produced from renewable energy, even though about 95% of hydrogen is currently produced from fossil fuels. The wind-based water electrolysis is selected for this environmental study as it demonstrates the lowest impact on climate change [51,52]. Finally, the infrastructure of the CO₂ capture and conversion processes as well as the hydrogen production process are considered in this study.

In a detailed review concerning the LCA of Carbon Capture and Utilization, von der Assen et al. [51] highlight that two impact indicators should always be included in the analysis: climate change (CC) and fossil resource depletion (FD). The direct CO₂ emissions, resulting from the process, and indirect CO₂ emissions due to utilities consumption are considered. In this study, five additional environmental impact indicators are also considered applying the standard practice ReciPe (H) midpoint approach [51,53]: terrestrial acidification (TA), fresh water eutrophication (FE), human toxicity (HT), water depletion (WD) and metal/mineral depletion (MD), in order to enhance the results and avoid misleading conclusions. More information about these impact indicators are presented in the Supplementary Information file (see SI.2.1).

Finally, the LCA Ecoinvent database [47] is used to provide the impacts of the conventional process. In order to compare the impacts of the conventional production of methanol with the alternative proposed in this paper, the data have been harmonized remodeling the inventory.

2.4. Uncertainty and sensitivity analysis

Qualitative and quantitative uncertainties are identified performing pedigree analysis and sensitivity analysis, respectively. The uncertainty analysis allows to analyze the uncertainty associated with the model output, while the sensitivity analysis helps to quantify how sensitive the output results are to variation of one or more variables. All the results are presented in the Supplementary Information file (see SI.3).

3. Description and modeling of the CO₂ capture and conversion processes

3.1. Simulation tools and process modeling

All the processes investigated in this paper were simulated in Aspen Plus® v10 software with the Redlich-Kwong equation of state [54] and Henry's law for the calculation of gaseous properties. For the calculation of liquid properties, the Electrolyte NRTL model [55–57] was used for the simulation of the CO₂ capture unit and the modified (Dortmund) UNIFAC model [58] for the simulation of the water-methanol distillation (CO₂ conversion unit). Graaf model [18] based on a dual-site Langmuir-Hinshelwood-Hougen-Watson mechanism [59] was considered to model reactions catalyst in the catalytic reactor (see SI.1.1 for details). The kinetic parameters of the model for a CuO/ZnO/Al₂O₃ industrial catalyst (45,776 – Copper based methanol synthesis catalyst from Alfa Aesar®) were determined from previous experimental data [60]. The Langmuir-Hinshelwood-Hougen-Watson model and its parameters are detailed in the Supplementary Information file (see SI.1.2).

3.2. Water electrolysis unit

Water electrolysis can be characterized in two different ways: the solution used as electrolyte (i.e. alkaline (AE), proton exchange membrane (PEME) or solid oxide (SOEC) electrolyzers) and the name of the source of electricity (i.e. high temperature steam, solar or wind electrolysis). As a result, the costs related to the production of hydrogen is highly related to the selected technique.

In the present work, alkaline water electrolysis was chosen as the main commercialized technique producing hydrogen (at 30 bar and 25 °C) [32] with a purity level of 99% with 80% hydrogen conversion efficiency [61]. The specific electrical power of these electrolyzers is estimated to 53.8 MWh per ton H₂ [32] and the investment costs to 638 €/kW_e [42] with an operation lifetime of 30 years. Oxygen is produced in high purity in the meantime and its selling was also considered as mentioned in part 2.2. Environmental considerations regarding this process are detailed in Supplementary Information file (see SI.2.2).

3.3. CO₂ capture unit

The CO₂ capture unit considered in this work was a CO₂ chemical absorption-regeneration process using amines as scrubbing solvent. The amine solution used for this absorption was an aqueous solution of monoethanolamine (MEA) in a mass concentration of 30 wt%, widely studied and considered as a benchmark for industrial CO₂ capture applications [32,36,62–64]. The flue gas to be treated comes from a conventional cement plant equipped with the Best Available Techniques (BAT) [65] and producing 3000 tons of clinker per day, which corresponds to a total flow rate of 250,000 m³/h flue gas (at 40 °C and 1.2 bar), with a CO₂ content of 20.4 mol% corresponding to a CO₂ flow of 2475 tons per day. Dedust, desulfurization and denitrification steps are already performed. The composition of actual flue gas from Norcem's cement plant in Brevik (Norway) was considered in this study [66] (see Table 1). The design of the CO₂ capture unit was based on the design of the CASTOR/CESAR pilot unit [67] that was scaled up to treat the flow rate coming from the BAT cement described before.

Fig. 1 shows a simplified process flow sheet of the CO₂ capture unit.

Prior to the CO₂ capture unit, the gas is preconditioned and its pressure is increased to 1.2 bar in order to compensate the pressure drop occurring in the absorber column. The CO₂ capture rate considered in the absorber column has been set to 90% and the CO₂ purity required at the top of the stripper unit to 98 mol%. The gas to treat (FLUEGAS) is then fed to the bottom of the absorber column (ABS). The chemical absorption process between CO₂ and MEA occurs in this absorption column, where the lean chemical solvent (LEANSOL), flowing from top to bottom, meets the flue gas flowing counter-currently. The rich amine solution (RICHSOL) is then pumped (PUMP-1) at 2 bar to the top of the stripper column (STRIP) through a lean/rich internal heat exchanger (HX) in which the rich solvent is heated to a temperature close to the stripper operating temperature. The stripper column includes a condenser and a reboiler made of a thermosiphon steam-driven system that supplies the heat required for the desorption and the regeneration of the amine solution. Finally, the CO₂ is released at the top of the stripper while the regenerated amine is recycled back into the absorber after being cooled through the internal heat exchanger (HX). A flash column (FLASH) is also added to condense the

Table 1
Molar composition of the gas to treat (G = 250,000 m³/h, 40 °C, 1.2 bar) [66].

Component	Concentration
N ₂	0.647
CO ₂	0.204
O ₂	0.086
H ₂ O	0.062
CO	1330 ppm
NO	474 ppm
SO ₂	111 ppm

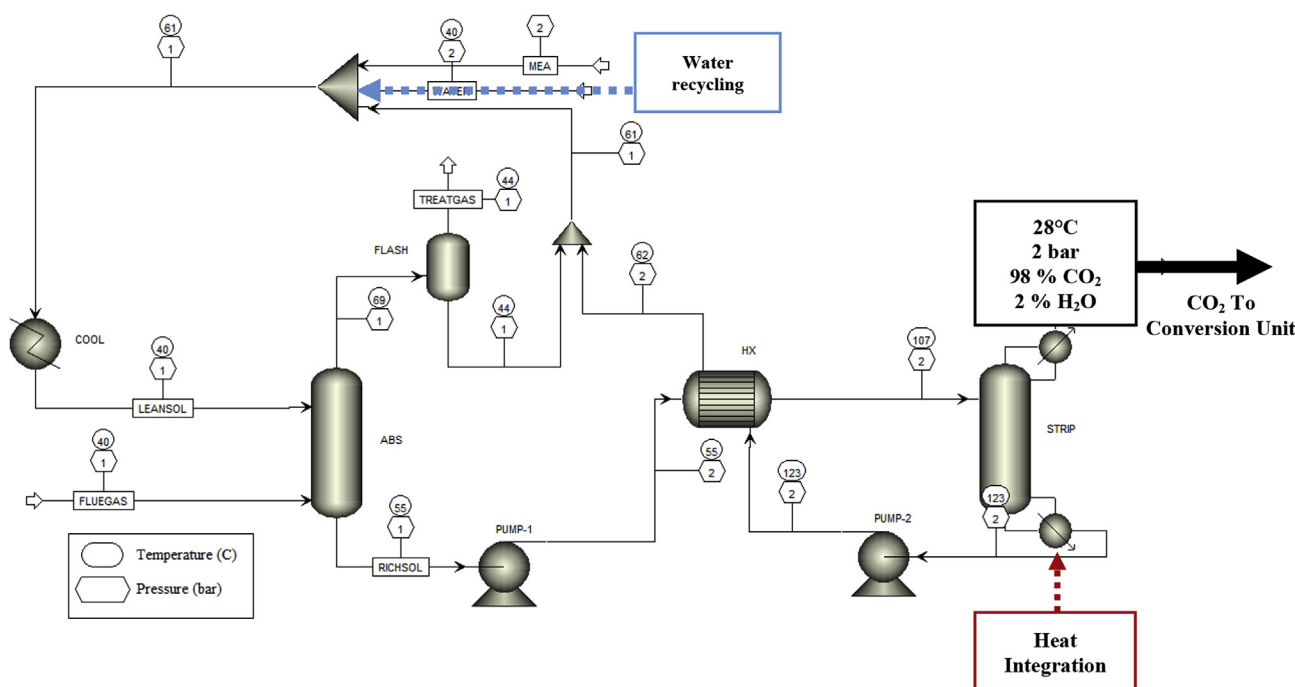


Fig. 1. Process flow sheet of the CO₂ capture unit.

vaporized fraction of MEA and water present in the treated gas coming out from the absorber. Moreover, water (WATER) and MEA (MEA) makeups are considered to compensate the loss of water through the system and the degradation of the amine solution respectively. The MEA makeup has been considered to 1 kg MEA per ton captured CO₂ to compensate the losses due to the thermal degradation of the amine only [32,68] as the evaporation of the amine in the absorber is prevented by the flash tank. The molar composition of the released treated gas is given in Table 2.

The main characteristics of the absorption and regeneration columns are given in Table 3.

Due to the excess heat available from the exothermal reactions occurring in the CO₂ conversion unit, heat integration is provided between the methanol conversion unit and the CO₂ capture unit. Moreover, water being produced as co-product of the methanol synthesis, water recycling is also proposed between the CO₂ capture and conversion units to reduce the water make-up for the regeneration of the solvent.

3.4. CO₂ conversion unit

The methanol conversion process proposed herein is based on the works performed by Van-Dal et al. [32] and Pérez-Fortes et al.

Table 2

Molar composition of the treated gas (201,460 m³/h, 44 °C, 1.2 bar).

Component	Concentration
N ₂	0.811
CO ₂	0.003
O ₂	0.107
H ₂ O	0.076
CO	1665 ppm
NO	648 ppm
SO ₂	117 ppm
MEA	0.457 ppm

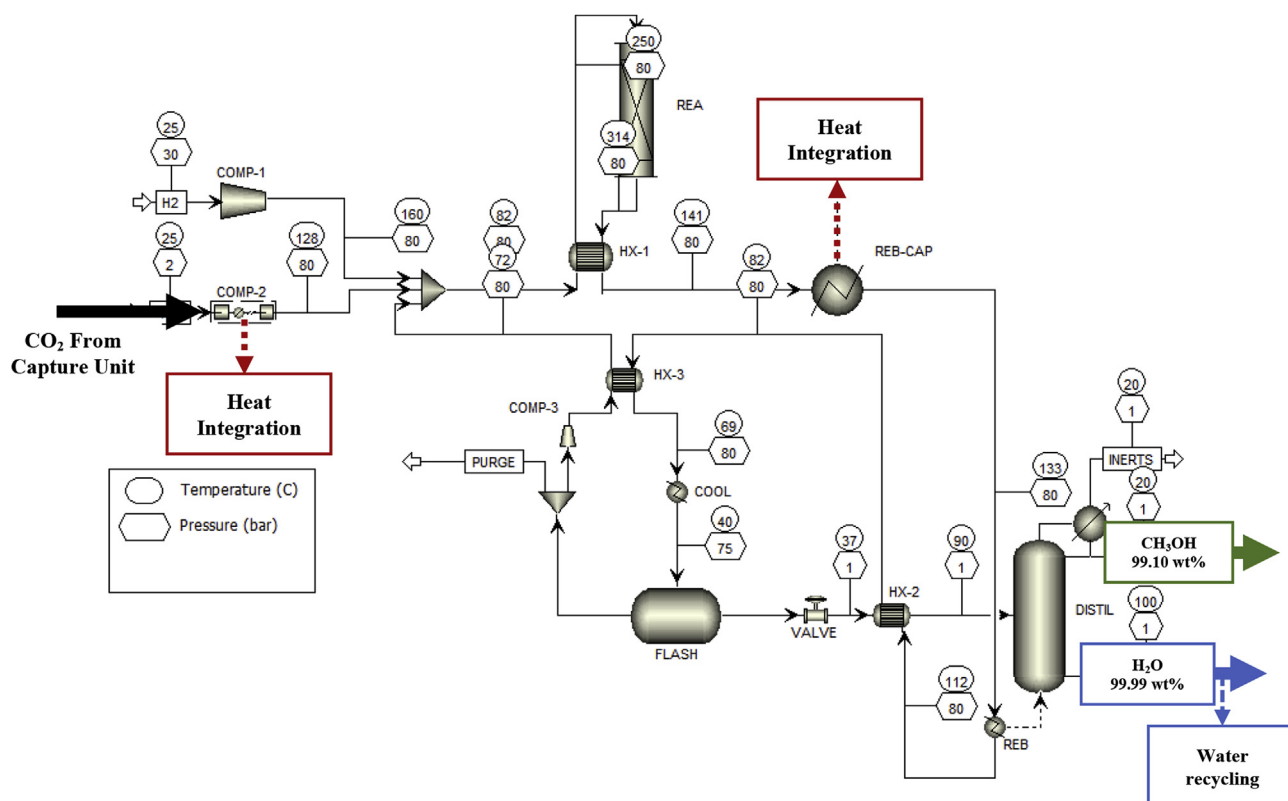
Table 3

Dimensions and operative conditions of the absorption and regeneration columns.

Specification	Absorber	Stripper	Unit
Diameter	8.7		m
Packing Height	17	10	m
Packing Type	50 mm – IMTP metal (Norton®)		/
Temperature (Feed)	40	107	°C
Pressure (Bottom)	1.2	2.0	bar

[26] regarding the design of the installation and scaled-up to convert the flow exiting the CO₂ capture unit and to produce pure methanol and pure water. The operative pressure has been set to 80 bar according to Fournel et al. [69] and the temperature at the inlet of the reactor to 250 °C to maximize the one-pass production of methanol. Fig. 2 shows the process flow sheet of the CO₂ conversion unit.

CO₂ coming from the CO₂ separation unit is fed at 2 bar with hydrogen coming at 30 bar, both at 25 °C. The hydrogen required for the conversion of CO₂ into methanol has been calculated considering a stoichiometric inlet H₂:CO₂ feed ratio of 3. Hydrogen is compressed to 80 bar in a single stage (COMP-1), where CO₂ is compressed to the same pressure in a series of four compressors (COMP-2) intercooled to 128 °C so that the recovered heat is provided to the reboiler of the stripper column from the CO₂ capture unit for heat integration. CO₂ and hydrogen are then mixed with the recycle stream derived from flash and preheated to 250 °C (HX-1) before feeding the adiabatic reactor (REA) where the methanol reactions occur. This adiabatic reactor is filled with 156 tons of a commercial CuO/ZnO/Al₂O₃ catalyst. The product stream leaves the reactor at 314 °C and is then thermally integrated by passing through a first heat exchanger (HX-1) to preheat the inlet gases of the reactor and reducing its temperature to 141 °C. A heat exchanger (REB-CAP) cooled then the stream to 112 °C and the heat recovered is provided to the reboiler of the CO₂ capture unit. The stream is further cooled in another heat exchanger (REB) to fully

Fig. 2. Process flow sheet of the CO₂ conversion unit.

provide the heat required by the distillation column for the water-methanol separation. The product stream is then further cooled in two heat exchangers to respectively preheat the liquid stream inlet of the distillation column (HX-2) and the unreacted gases recycled to the adiabatic reactor (HX-3). At this point, the temperature of the product stream is 63 °C and a cooler (COOL) is required to cool it down to 40 °C to be fed to a flash tank (FLASH) for the separation of the gaseous and liquid phase of the stream. The pressure of the flash tank has been set to 75 bar to consider the pressure drop occurring along the line. The unreacted gases are then recycled to the inlet of the adiabatic reactor but a fraction of them are purged (PURGE) to prevent the accumulation of inert gases and by-products in the reaction loop.

The water-methanol liquid mixture leaving the flash tank is then expanded to 1 bar and preheated (HX-2) to 90 °C before being sent to the distillation column. The features of the distillation column depend on the desired purity/grade. Thus, to reach the design specifications of methanol and water purities, pure liquid methanol (99.1 wt%) is produced at 1 bar and 20 °C at the head of the distillation column, where pure water (99.99 wt%) is recovered at 1 bar and 100 °C at its bottom.

Considering the CO₂ capture and conversion units as an overall process, an external heat integration was investigated. As the result, the external heat exchanger (REB-CAP) set between HX-1 and the reboiler of the distillation unit lead to a 3.8 MW that can be brought to the reboiler of the CO₂ capture unit. Moreover, the heat recovered at 128 °C from the compression of the inlet CO₂ through COMP-2 can also be recycled to the reboiler of the CO₂ capture unit, accounting for a 8.6 MW.

Table 4 presents data of the CO₂ conversion unit configured to convert 2227 tons CO₂ per day into 1546 tons methanol per day.

Table 4

Main process data equipment for the CO₂ conversion unit.

Equipment	Specification	Unit
Reactor	Inlet temperature	250 °C
	Pressure	80 bar
	Reactor volume	120 m ³
	Catalyst type	CuO/ZnO/Al ₂ O ₃ /
	Mass of catalyst	156 ton
	Apparent density	1,300 kg/m ³
Distillation column	Diameter	5.1 m
	Tray spacing	0.61 m
	Tray type	Bubble-cap /
	Number of plates	28 (+2) /
	Feed-stage location	19 /
	Temperature (feed)	90 °C
	Pressure (top)	1 bar

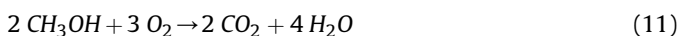
3.5. Valorization of inert and purge gases

To fully optimize the CO₂ conversion process, it is mandatory to consider the possible reuse of undesired flows such as the inert stream from the head of the reboiler column and the one exiting from the purge, especially when these streams contain significant amount of hydrogen. Indeed, as these streams are quite rich in hydrogen, CO and methanol, their combustion to produce reusable steam seems to be an attractive solution. Considering the combustion of these exhaust gases, the combustion reactions considered are the following (see Eqs. (9)–(11)) and the Lower Heating Values (LHV) are summarized in Table 5.



Table 5
Lower Heating Values (LHV) of combustible gases from exhaust streams [32].

Component	LHV	Unit
H ₂	121.0	MJ/kg
CO	10.1	
CH ₃ OH	19.9	



The partial mass flow rates of the inert and purge gases recovered from the CO₂ conversion process are presented in Table 6.

Considering a 85% efficiency in the boiler used for the combustion of these gases, the heat recovery from this stream generation was calculated to reach 17.6 MW, that could be recycled to the reboiler of the CO₂ capture unit to reduce its dedicated heat demand. This heat integration is of importance as it exceeds alone the global heat integration from COMP-2 and REB-CAP.

As a result, the overall heat integration between the CO₂ capture and conversion units can reach up to 32.7% (i.e. 30.4 MW) by considering the heat integrations from COMP-2, REB-CAP and the combustion of exhaust gases from the CO₂ conversion process (see Table 7).

4. Results and discussion

4.1. Technological assessment

The technological indicators of the CO₂ capture and conversion processes are presented in Tables 7 and 8.

The presented process is able to treat 2475 tons of CO₂ per day and convert 90% of them into 1546 tons of methanol. The CO₂ absorption rate of the CO₂ capture unit is 90% and the external heat duty to provide to reboiler for the regeneration of the solvent can be reduced from 3.7 GJ/ton captured CO₂, which is within the range of a conventional 3.6–3.9 GJ/ton captured CO₂ [70], to 2.7 GJ/ton captured CO₂, considering the heat recovered from the conversion unit. This reduction of the stripper reboiler duty corresponds to a direct 32.7% of the total energy requirements for the regeneration of the amine solution. The use of the flash allowed the recirculation of the vaporized MEA fraction and the reduction of the water makeup. As a result, the water from the coproduct of the methanol synthesis completely fulfils the water make-up from the CO₂ capture unit and represents only 2.5% of the water produced by the CO₂ conversion unit. The CO_x conversion per pass in the reactor is 20.1% and the overall conversion of the process is 93.4%. The reduction of this conversion yield by 6.6% is directly related to the purge ratio of the process set to 0.5% to prevent the accumulation of inert gases like nitrogen and being thus responsible for a direct 1.9%-loss of the

Table 6
Partial mass flow rates of the inert and purge gases from the CO₂ conversion process.

Component	Stream		Unit
	Purge	Inert	
CO ₂	1020.2	1809.6	kg/h
H ₂	481.3	36.0	
CO	474.5	59.3	
CH ₃ OH	30.6	290.6	
N ₂	8.6	0.8	
H ₂ O	3.9	0.3	

Table 7
Key technological indicators for the CO₂ capture process.

Indicator		Specification		Unit
Mass indicators	CO ₂ inlet flow	2,475	ton/day	
	CO ₂ production flow	2,227	ton/day	
	L/G ratio (absorber)	0.0059	m ³ /m ³	
	Recycled water make-up	21.5	ton/day	
	MEA make-up	2.23	ton/day	
Energy indicators	Cumulative power of pumps	0.43	MW	
	Reboiler duty (before integration)	93.2	MW	
		3.7	GJ/tCO ₂	
	Reboiler duty (after integration)	62.8	MW	
		2.7	GJ/tCO ₂	
Energy savings		32.7	%	

inlet CO₂ flow rate. Other losses of CO/CO₂ compounds, which are not converted in the reactor, come from their respective solubility and volatility in the methanol (1.8%) and head (2.9%) streams of the distillation column. The proposed process shows significant integrations, allowing to reduce the reboiler duty of the distillation column by 100% through internal heat exchangers.

Fig. 3 presents the integrations between the CO₂ capture and conversion units and summarizes the technological assessment of the CO₂ capture and conversion units normalized to the production of one-ton methanol to be comparable to the conventional production from syngas.

4.2. Economic assessment

The global CAPEX of the CO₂ capture and conversion processes is estimated to 47.3 M€ (without considering the CAPEX of the electrolyzers). The repartition of this global CAPEX is presented in Fig. 4. The CAPEX related to the equipment of the CO₂ capture and conversion processes is estimated to 18.4 M€ with about 37% and 63% respectively dedicated to the CO₂ capture and conversion units. Other major contributions comprise contingencies and piping that account for 15% and 10% respectively.

The repartition of the CAPEX dedicated the purchase of the equipment is presented in Fig. 5. The most expensive equipment is the compressors, which contribute for 41% of the overall equipment CAPEX of the process. Heat exchangers, reactor and columns contribute respectively for 29%, 4% and 23% of the above-mentioned CAPEX. The costs of flash tanks are marginal with only 3% contribution to the equipment CAPEX.

Considering the assumptions exposed in the economic indicators, the OPEX of the CO₂ capture and conversion unit (including the production of H₂) is presented in Fig. 6. The globalized OPEX dedicated to the CO₂ capture and conversion are calculated to reach 31 € and 23 € per ton methanol respectively. Calculated per ton of captured CO₂, the OPEX of the CO₂ capture and conversion units are estimated to 22 € and 17 € respectively.

Based on these values, the economic benefits of the CO₂ capture unit is uncontroversial compared to the OPEX of 33 € per ton CO₂ captured [71] with a benchmark MEA-scrubbing process. The 33.3% reduction in the OPEX can be directly linked to the 32.7% energy savings dedicated to the thermal integration between the CO₂ capture and conversion unit. The amortization of the CAPEX is also estimated to 8 € per ton methanol.

The selling of methanol remains the main economic benefit of the process with a selling price of 450 € per ton methanol. The selling of oxygen co-produced with the electrolysis of water and the current value of CO₂ credit tax also improves the economic balance of the overall process. Their contribution represents together 20% of the global revenue of the installation. In the current situation, the integrated CO₂ capture and conversion process is still not

Table 8
Key technological indicators for the CO₂ conversion process.

Indicator	Specification	Unit
Mass indicators	CO ₂ inlet flow	2,227 ton/day
	Hydrogen needs	306 ton/day
	Oxygen co-production	2,449 ton/day
	Methanol production	1,546 ton/day
	Water production (recycle incl.)	895 ton/day
	Recycled water flow	21.5 (2.5% of produced water) ton/day
	Purge split ratio	0.5 %
	H ₂ :CO ₂ ratio (feed/reactor inlet)	3 / 5.3 /
	CO ₂ conversion per pass	20.1 %
	Methanol conversion yield	93.4 %
	Reflux ratio (distillation)	1.52 /
Energy indicators	Cumulative power of compressors	21.3 MW
	Condenser (distillation) temp.	20 °C
	Condenser (distillation) duty	-57.8 MW
	Reboiler heat duty	0 MW
	Cooler heat temp.	40 °C
	Cooler heat duty	-22.4 MW

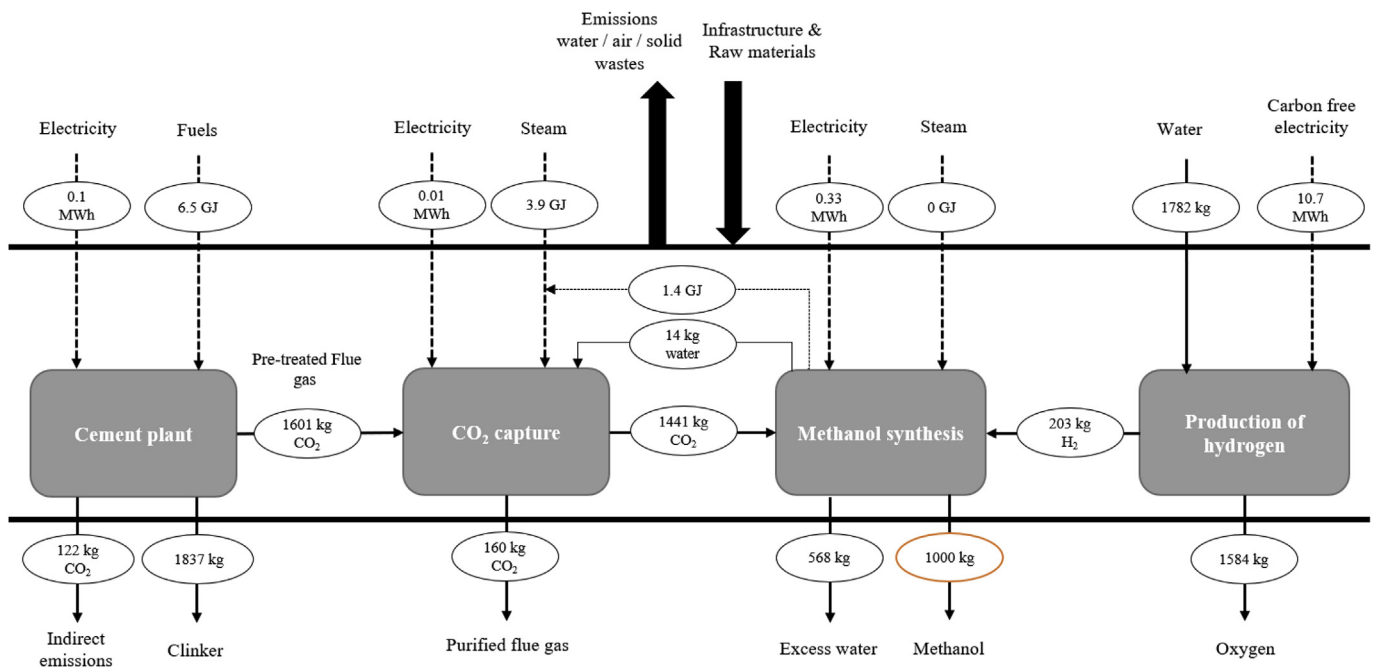


Fig. 3. Technological metrics of the CO₂ capture and conversion units normalized to the production of one-ton methanol.

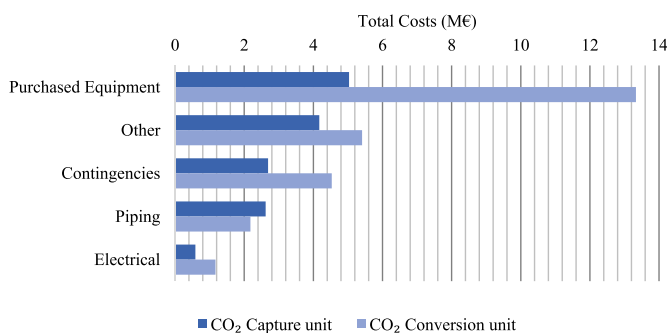


Fig. 4. Total capital investment costs of the CO₂ capture and conversion units.

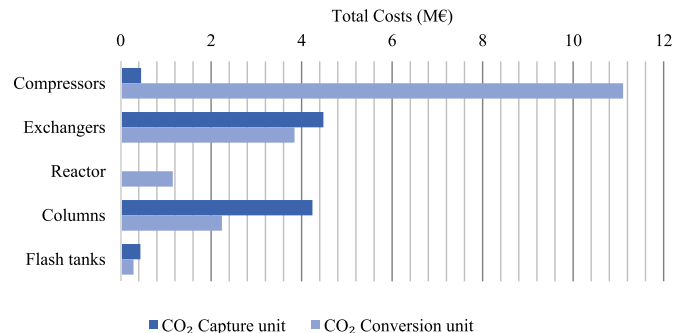


Fig. 5. Total installed equipment costs of the CO₂ capture and conversion units.

economically viable because the main cost-effective unit of the process remains the production of hydrogen, directly related to the

electricity price of 70 €/MWh [33]. The total costs of production (i.e. 808 € per ton methanol) are 1.5 times the expected revenue

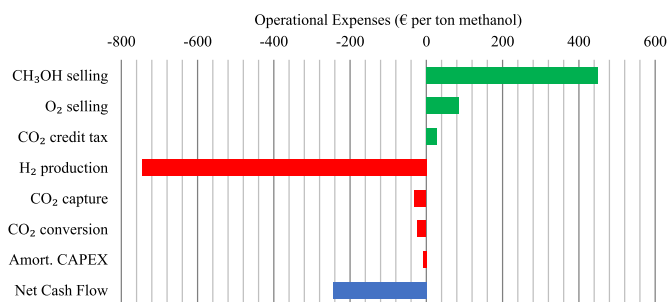


Fig. 6. Operational costs of the CO₂ capture and conversion units.

from methanol, oxygen and carbon credits selling (i.e. 556 € per ton methanol).

However, to be completely rigorous regarding the financial analysis of the CO₂ capture and conversion process, the CAPEX related to the electrolyzers of the hydrogen production has to be considered. This relative CAPEX is quite expensive as it can be approximated to 638 €/kW [42]. Considering that the power of the electrolyzers required to produce the hydrogen used in the CO₂ conversion process (i.e. 306 tons H₂ per day) is close to 687 MW, the CAPEX could be thus approximated to 438 M€. This huge CAPEX value dedicated to the hydrogen production, representing more than nine times the CAPEX of the combined CO₂ capture and conversion processes, has also been confirmed by Pérez-Fortes et al. [42] and has a dramatic influence on the evaluation of the NPV of the overall process. As a result, based the above-mentioned assumptions, the NPV's of the CO₂ capture and conversion process have been estimated to -920 M€ and -1358 M€ with and without considering the CAPEX of the electrolyzers respectively. These high negative values clearly demonstrate that the described process is currently economically unprofitable on a 10-year horizon. However, it is important to qualify this statement as the calculation of NPV is based on many fluctuating values and especially the costs of electricity and the price of methanol considered as assumptions. To evaluate the potential profitability of the process, a bivariate analysis was performed to quantify the influence of methanol selling prices and electricity cost on the NPV of the process.

Bivariate sensitivity analyses – Bivariate (or multivariate) analyses are generally considered to give more realistic value of the breakeven prices (to reach NPV equal to zero). In this perspective, the bivariate sensitivity of the electricity price versus the methanol selling price has been studied as both these values had the most important contributions on the NPV. The results of this analysis are presented in Fig. 7, which presents trade-off values to consider to

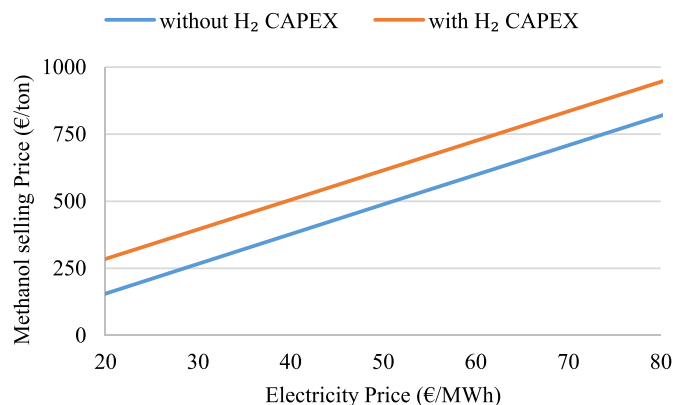


Fig. 7. Bivariate analysis of the electricity and methanol price (NPV = 0).

have NPV equal to zero for the overall process. As a result, in place of requiring an electricity price lower than 46 or 34 €/MWh (considering or not the H₂ CAPEX) or a methanol selling price higher than 716 or 843 €/ton, other couples of breakeven prices can be considered. A realistic example of these couples could be an electricity price of 50 €/MWh related to a methanol selling price of 496 €/ton (without the H₂ CAPEX).

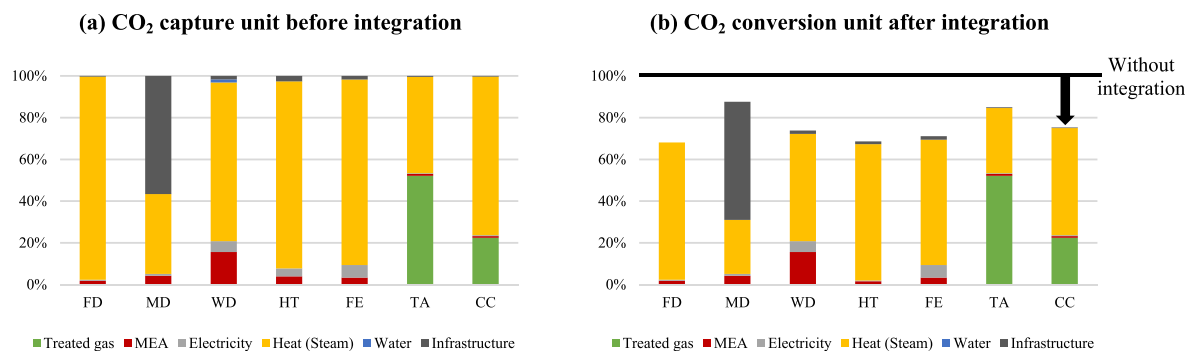
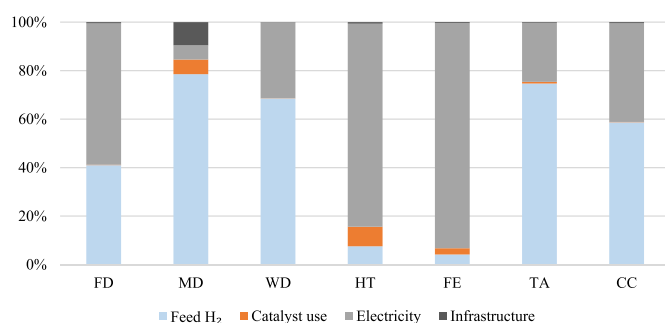
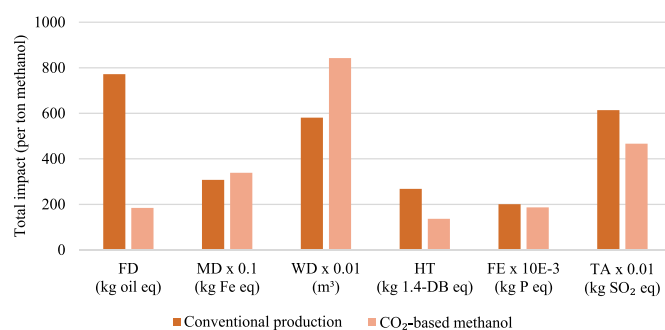
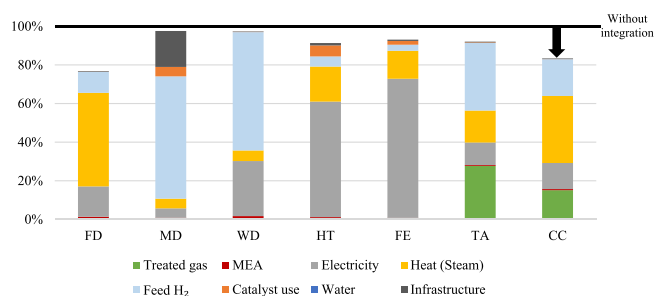
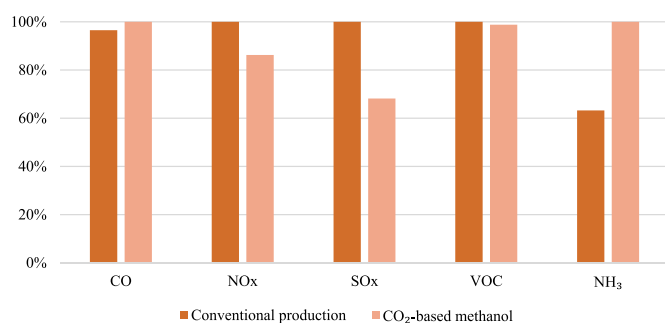
4.3. Environmental assessment

The life cycle assessment is conducted on seven relevant impacts indicators. Fig. 8 presents the contribution analysis of the CO₂ capture unit. As it can be observed in Fig. 8-a, the heat demand for the regeneration of the MEA solvent contributes to 76% and 97% of the CC and FD, respectively. As a result, it appears to be the most impacting demand in comparison to the use of electricity, water consumption and MEA production. The contribution of MEA use related to the MEA makeup appears to be negligible except for WD (where it represents 16% of the impact factor). However, the use of this solvent still presents high-energy penalty for its regeneration and could also degrade the equipment due to corrosion issues. As a result, the use of alternative configurations or solvents [72] could be an opportunity to lower these impacts but their current TRL are currently too low for industrial applications. Finally, even if the major part of CO₂ has been captured through the process, the remaining treated gas from the head of the absorber still has an environmental impact and contributes to 52% of TA. Fig. 8-b presents the effect of the integration of the CO₂ capture and conversion processes on the impacts of the CO₂ capture unit alone. Major reductions can be observed for WD (26%), FD (32%) and CC (25%) thus demonstrating the environmental benefits of the integration.

Fig. 9 summarizes the environmental assessment of the CO₂ conversion process. In this process, hydrogen supply (Feed H₂) is one of the main contributors to most impact categories and represents 59% of CC, 41% of FD and almost 70% of WD. The high impacts on MD are explained by the use of various mineral resource for the infrastructure construction of the electrolyzer. As the wind-based water electrolysis was selected for the hydrogen supply, it is difficult to further reduce the environmental impacts related to the source of electricity for the electrolysis. Indeed, it has no meaning to produce methanol from CO₂ if hydrogen is supplied by steam-methane-reforming (as it is currently the case). The catalyst contribution is usually negligible, its major impacts contributing for 8% and 6% of HT and MD respectively. The compression of H₂ and CO₂ inlets to reach the working pressure of 80 bar also contributes to FE (90%), HT (84%), FD (60%) and CC (40%). Finally, the impacts of the inert and purge gases being largely outweighed by the impacts of the other contributing categories, they are considered as negligible.

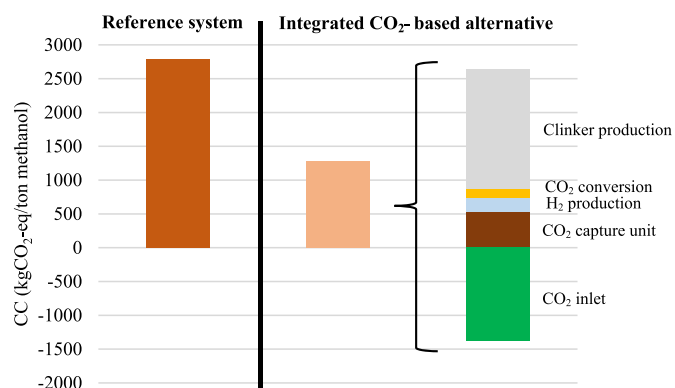
Fig. 10 demonstrates the environmental benefits of the water and heat integration between the CO₂ capture and conversion processes. These integrations are positive on all the environmental impacts, which show a global reduction of 2%–24% according to the considered criterion. Particularly, the environmental impacts on CC and FD are reduced by 17% and 23% respectively.

Non-CO₂ emissions (CO, NO_x, SO_x, VOC, and NH₃) are also evaluated to compare the conventional production of methanol (data available in the EcoInvent [47] v3 database, see SI.2.3) from syngas [73,74] and the CO₂-based alternative presented in this work. Results are compared per ton methanol and then normalized to the largest emission value (see Fig. 11). The CO and VOC emissions are similar, while the NO_x and SO_x emissions are lower for the CO₂-based production of methanol compared to the conventional production. However, higher NH₃ emissions are observed for the alternative methanol production, mainly due to the use of MEA-

Fig. 8. Contribution analysis of the CO₂ capture unit alone.Fig. 9. Environmental assessment of the CO₂ conversion process.Fig. 12. Comparison between the environmental impact factors of the integrated CO₂ conversion process with the conventional production of methanol (from syngas).Fig. 10. Environmental assessment of the integrated CO₂ capture and conversion process.Fig. 11. Comparison between non-CO₂ emissions of the integrated CO₂ conversion process with the conventional production of methanol (from syngas).

solvent to capture the CO₂.

Finally, Fig. 12 presents the comparison between the environmental impacts of the conventional production of methanol and the CO₂-based alternative.

Fig. 13. Comparison between the Climate Change impact (CC) of the integrated CO₂ conversion process with the conventional production of methanol (from syngas).

The environmental impacts of the CO₂-based process are mostly lower than the conventional methanol production especially analyzing the FD that is reduced by 77% in comparison to the conventional process. Exceptions remain for MD and WD, which can be explained by the production of hydrogen from water electrolysis. Regarding CC, it demonstrates a reduction (see Fig. 13): 1300 kgCO₂-eq per ton methanol for the integrated CO₂-based process in comparison to 2800 kgCO₂-eq per ton methanol for the reference system based on the CO₂ emissions from cement plant without CO₂ capture and from the conventional process for the production of methanol from steam reforming of natural gas. The CO₂ emissions of the CO₂-based process are thus reduced by over 50% in comparison to the reference system.

5. Conclusions

In this work, the entire CO₂ conversion chain from its capture to its conversion into methanol was investigated. In this perspective, the CO₂ conversion process has been optimized with regards to the operative conditions and considering the internal and external possible integrations. The valorization of the residual exhaust gases of the process has also been proposed.

The integrated and optimized process is able to treat 2475 tons per day CO₂ and convert 90% of them into 1546 tons methanol per day. The heat integration inside the conversion unit reduces the reboiler duty required for the distillation of the water-methanol mixture by 100%. The heat integration between the CO₂ capture and conversion units also reduces the reboiler duty required for the regeneration of the solvent by 32.7% in comparison to conventional MEA absorption processes. The water recycled from the coproduct of the methanol synthesis completely fulfils the water demand of the CO₂ capture unit for water make-up. The global CAPEX of the installation has been estimated to 47.3 M€ with 18.4 M€ dedicated to the installed equipment costs without considering the CAPEX related to the hydrogen production. Considering a current price of electricity of 70 €/MWh, the process is still not economically viable with the total cost of production being almost 1.5 times the expected revenue. However, its economic breakthrough occurs for an electricity price of 46 €/MWh or 34 €/MWh (considering or not the H₂ CAPEX), which is already considered as realistic in other studies or available in some countries like Iceland.

The environmental assessment of the CO₂-to-methanol value chain highlights that hydrogen supply is the main contributor in most impacts' categories, even if hydrogen is produced from renewable energy. Although the contribution of the production of MEA solvent appears to be negligible, the energy penalty due to its regeneration is significant to most impacts. In addition, the released treated gas still contributes to the terrestrial and climate change impacts. The comparison with the conventional production of methanol from steam methane reforming highlights the potential of this alternative route regarding mainly fossil depletion. Its contribution to climate change is drastically lower than the conventional production as the climate change impact of the integrated process is reduced by over 50% in comparison to the emissions related to the conventional production of methanol from natural gas. As a result, the proposed CO₂-based conversion process can be considered as a credible substitution alternative for methanol production and an interesting opportunity in the global picture of climate change mitigation, which is close to a real project situation as long as transportation is avoided.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.renene.2019.07.010>.

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